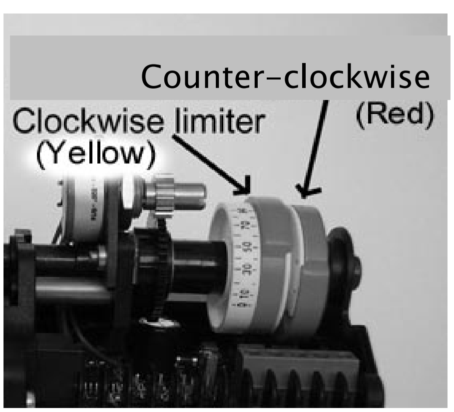
For each motor, it is necessary to adjust the servo-motor position relative to the shaft it is driving (e.g. gas valve) in order to obtain the correct open and closed positions on the display.
To do this, follow the procedure below:

1. **Ensure that the correct servo-motor direction is set before connecting the servo-motor to the relevant valve.** If the servo-motor direction is incorrect use the relevant option parameter to reverse.
2. Move the valve to its fully closed position and adjust the servo-motor position by driving the motor so that approximately 1° is shown on the display.
3. Move the valve to its fully open position by driving the servo-motor and check that the display reads approximately 90° or the maximum angular opening required from the servo-motor if this is less than 90°.

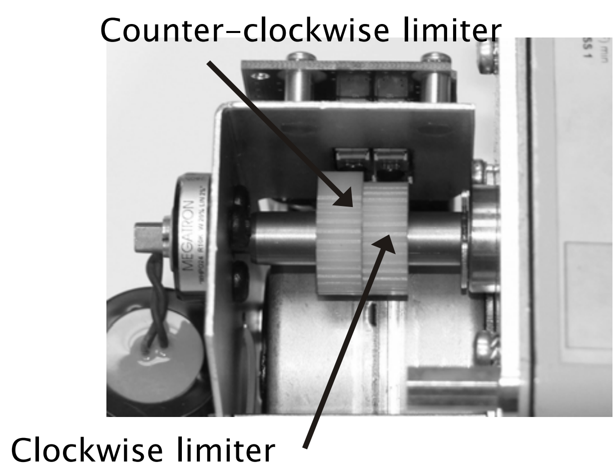
3.5 Adjusting microswitch positions

Each time a burner start-up sequence is initiated, the control will move the fuel and air damper motors to their respective closed positions to prove correct motor and potentiometer operation. Each motor has microswitches fitted to set the close position obtainable during this proving operation, and limit the maximum 'open' position to prevent burner/boiler damage in the event of a system failure.
To set the microswitch positions, follow the procedure below.

1. Enter commission ratio mode (see section 5)
2. Move each motor to approximately 45°, using the **UP/DOWN** keys, this is to ensure the **DOWN** key will drive the motor.
3. Holding the **DOWN** key, tighten up the low limit microswitch until the motor will no longer move down.
4. Holding the **DOWN** key, gradually slacken off the low limit microswitch until the motor starts moving down. Continue to slacken off the microswitch until the motor stops with a reading on the display of approximately 1°.
5. Move the motor up and down a few times to check that the motor stops each time at approximately 1°, and re-adjust the microswitch if necessary. This position will allow for some tolerance in microswitch operation.
6. Hold the **UP** key and tighten up the high limit microswitch until the motor will no longer move up.
7. Holding the **UP** key, gradually slacken off the high limit microswitch until the motor starts moving up. Continue to slacken off the microswitch until the motor stops in the desired purge position. This position does not have to be 90°, but it is recommended that it is more than 45° and less than 90°.
8. Move the motor, up and down a few times to check that the motor stops each time at the desired 'limit' position. Repeat steps 3 to 7 if necessary.



NOTE: For NXC04, NXC12, NXC20 SERVOS ONLY



NOTE: For NXC40 SERVO ONLY

3.6 Servomotor Replacement

After a system has been commissioned replacing an undersized or failed servomotor requires the following considerations:

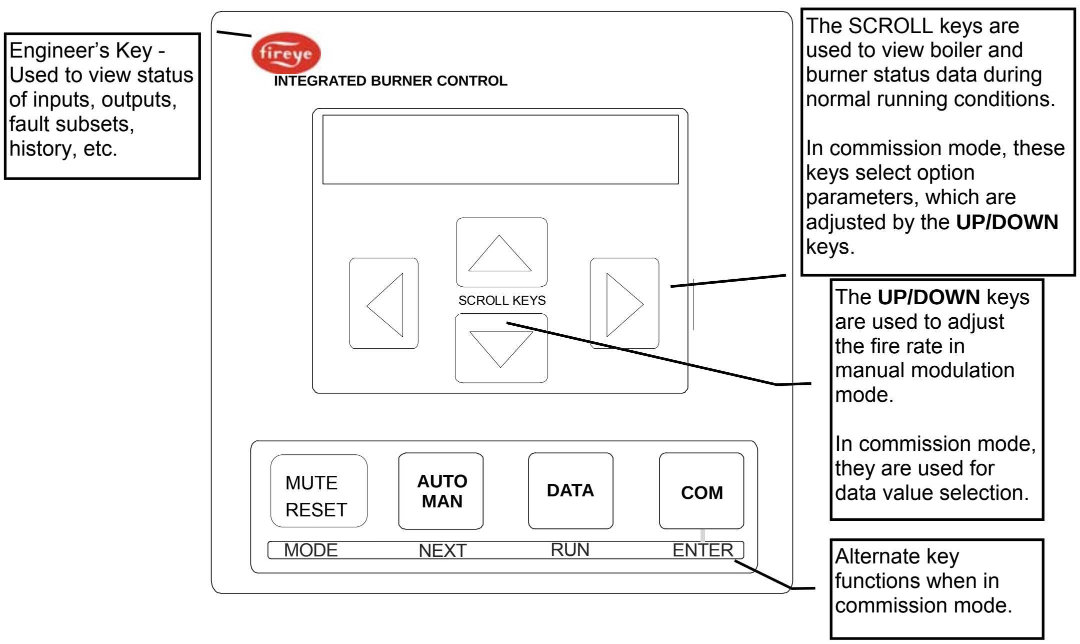
1. First determine the direction the motor travels as the replacement will have to be set the same way. This can be found in Option Parameter 5.x (x being the number of the drive. Eg. 5.3 is drive 3). If the device being driven rotates clockwise to increase firing rate, then the servomotor must drive counter clockwise (as viewed from the servomotor shaft) and vise versa.
2. **DO NOT COUPLE THE MOTOR TO THE DRIVEN SHAFT UNTIL AFTER THE DIRECTION HAS BEEN SET. SEE SECTION 3 OF THIS MANUAL.**
3. Deselect the original servomotor serial number using Option Parameter 3.x.
4. Select the new servomotor serial number from Option 3.x. The serial number appears on the servomotor label but will also appear as “unreserved” in the list.
5. The burner will have to be started in full commissioning mode and each position, P0 though PX (high fire), must be verified by using the “next” key on the display. This is covered in the commissioning section of the manual (Bulletin PPC-6001). Combustion should be checked while doing this so as to assure a safe operation.

4 Description of operation

4.1 The Display / Keypad

The display is a 2 line, 20 character per line, dot matrix vacuum fluorescent type allowing the use of plain text messages for most display parameters. The keypad is a membrane construction with tactile keys to give a positive feedback of the actuation.

NOTE: The display can be attached to more than one PPC6000 via CANbus. The display must be interrogating the address of the desired PPC6000, if not a fault message will be displayed. See the troubleshooting section (Section 6.7.1) for details.



NOTE: Pressing both the left and right scroll keys simultaneously will reveal the serial number of the display, as well as the status of a backup. The letters CDBKP will appear on the bottom line if a valid backup is stored.

Key	Function
fireye	Selects Engineers Key mode. See section 6.6 for further details.
MUTE/RESET	Press this key to mute (open) the alarm relays, then hold the key down for three seconds to remove any cleared faults.
AUTO MAN	Selects auto or manual burner modulation. In manual mode the **UP/DOWN** keys are used to alter the firing rate.
DATA	Toggles the display between normal run mode and servo-motor position. In **COM**missioning mode this key is used to return to run mode. Selects different data types on the display window. Holding the DATA key down for 5 seconds allows for selection of operating modes: NORM = normal / remote, used for sequencing. Loc1 = use PID 1 only Loc2 = use PID 2 only LEAD = Boiler is Lead (upper case) lead = Boiler is not lead (lower case) OFF = Turns burner off
COM	Changes operation to commission mode via a passcode. In **COM**missioning mode this key is the "enter" key.

4.2 Start-up sequence

This fuel / air ratio control performs burner start-up and shut-down in conjunction with an external 'burner controller'. The external 'burner controller' provides burner management functions such as flame / air pressure monitoring and it also controls the fuel shut-off valves and combustion air fan.

The start-up and shutdown sequence is handled by a progression of stages, each requiring a certain set of conditions to move on to the next one. The progression through the stages requires 'handshaking' between the two devices. This is accomplished with a variety of signals. Relevant Engineers Key, if available, will be shown as (EK#).

From burner controller to fuel-air ratio controller:

Signal Name:	Alternate Names / Pin:	Description:
H (HIGH) (EK5)	Purge request, PA9. Low Voltage	Commands fuel / air ratio controller to move the servos UP for either a pre-purge or a post purge. LOW VOLTAGE. This input is made by connection to PA11. If this input comes on during burner normal run (modulation), the burner will be turned off and a post-purge initiated.
A (AUTO) (EK6)	Release to Modulate, PA10. Low Voltage	Releases the fuel / air ratio controller to modulate as required to support the load on the boiler. LOW VOLTAGE. This input is made by connection to PA11. If this input goes OFF during a normal run, the motors will move to their low-fire (not ignition) positions and stay there. No feedback will be given on LFS (see below). (See OPT 16.1 for return to pilot)
PROFILE SELECT (EK11-14) (EK31)	PE9, PE10, PE11, PE12 Line Voltage	Tells the fuel / air ratio controller which fuel / air profile to run. This signal must be removed when the burner goes off, before a new start-up can begin. It is common to use a selector switch fed from the fan contactor output of the burner controller for this input. LINE VOLTAGE. If this signal is removed at any time, the controlled shutdown output (PE3) will turn off for at least 3 seconds to ensure the burner is off. (See OPT 16.2 for profile swap on the fly.)

From fuel-air ratio controller to burner controller:

Signal Name:	Alternate Names / Pin:	Description:
SAFETY SHUTDOWN	Lockout, PE5 – PE6. Line Voltage	These contacts will open in the event of a lockout of the fuel-air ratio controller. When this happens, PE3 will loose power a short time later. These contacts MUST be in the main safety circuit of the Flame Safeguard, effectively interrupting power to the fuel valves. LINE VOLTAGE. The installation must guarantee that if these contacts open, the burner goes off IMMEDIATELY.
CONTROLLED SHUTDOWN (EK18) (EK30)	Call for heat. PE3. Line Voltage	When the fuel-air ratio controller requires the burner to come on, LINE VOLTAGE will be present on this terminal.

Signal Name:	Alternate Names / Pin:	Description:
HFS (HIGH FIRE SWITCH)	Purge proved, PE8. Line Voltage	When the purge (P1) position is reached (pre-purge or post-purge), LINE VOLTAGE will be present on this terminal. Line voltage will also be present here when the control is modulating and high fire is reached, but only if the AUTO input is ON. (See OPT 16.2 for profile swap on the fly.)
LFS (LOW FIRE SWITCH)	Ignition proved, PE7. Line Voltage	When the ignition (P2) position is reached and the fuel-air ratio controller is ready for ignition, LINE VOLTAGE will be present on this terminal. Line voltage will also be present here when the control is modulating and low fire is reached, but only if the AUTO input is ON. . (See OPT 16.2 for profile swap on the fly.)

The startup / shutdown stages are as follows:

Stage no.	Stage name	Description
0.	Non-volatile lockout / safety shutdown	The burner is held in this state until all faults are removed. The 'safety shutdown' output (PE5 – PE6) will be open. The 'controlled shutdown' output on PE3 will also be OFF during this time, however if the burner was running when the fault occurred, the 'safety shutdown' output will have opened first.
1.	Burner off (EK18)	The burner is checked to make sure that it has switched off completely. The fuel/profile select inputs must all go OFF when the burner is switched off (or at the end of post-purge). This provides a feedback to confirm that the burner is off, and ensures that this control is always synchronized with the burner controller. The 'controlled shutdown' output on PE3 will be OFF during this time, however the 'safety shutdown' output PE5 – PE6 will be closed during this time. The control will advance to status 2 when: - All fuel/profile select inputs are OFF - The 'boiler status' is equal to 1 (call for heat). See engineers key 18. - The 'AUTO' input (PA10 – PA11) is OFF (open).

Stage no.	Stage name	Description
2.	Wait for purge (EK5) (EK11-14) (EK31)	The ‘controlled shutdown’ output on PE3 will be switched on, to tell the burner controller there is a call for heat. This control waits for a fuel/profile select signal on one of the (line voltage) terminals PE9, PE10, PE11 or PE12 and a purge request (HIGH input, PA9 - PA11 closed. This input is low voltage). This would normally come from the burner controller. If the fuel/profile select input is removed (PE9, 10, 11 or 12), the control will immediately move to status 1. Note: Statuses 3 and 4 are reserved for a gas valve proving system and are not implemented on this control. This control will advance directly to status 5 once the conditions above have been met.
3. & 4.	N/A	Not Applicable to PPC6000
5.	Prove closed positions (EK80-89)	In all profiles, the fuel and air motors are moved down until they are stopped by the ‘closed position’ micro-switches in the servos. The final positions are compared with the closed positions stored in memory, and must be within ±5° of the stored values otherwise the control will lockout. Variable speed drives must read zero. (4 mA). This includes any VSD (VFD) assigned to the chosen profile. When all drives have stopped moving, the control will advance to status 7. The ‘HIGH’ input (PA9 to PA11) still must be made during this time. If the fuel/profile select input is removed (PE9, 10, 11 or 12), the control will immediately move to status 1. Note: Status 6 is reserved for air pressure detection, and is not implemented in this control. This function must be provided by the burner controller.
6.	N/A	Not Applicable to PPC6000
7.	Moving to Purge	The selected motors are moved up towards the purge position. When the drives have all stopped, the control moves to status 8. If the fuel/profile select input is removed (PE9, 10, 11 or 12), the control will immediately move to status 1.

Stage no.	Stage name	Description
8.	Pre-purge	The controller confirms all required drives are at their purge positions, and gives a 'purge proved' signal by providing (line voltage) to terminal PE8 (high-fire-switch). The control will remain in this status until the burner controller signals the end of pre-purge by breaking PA11 – PA9 ('HIGH' input). If the fuel/profile select input is removed (PE9, 10, 11 or 12), the control will immediately move to status 1.
9.	Move to ignition positions	The selected motors are moved to their ignition positions. The 'purge proved' signal is switched off. When the drives have all stopped moving, the control advances to status 10. If the fuel/profile select input is removed (PE9, 10, 11 or 12), the control will immediately move to status 1.
10.	Ignition	The controller confirms all required drives are at their ignition positions, and gives an 'ignition proved' signal by supplying line voltage to terminal PE7 (low-fire-switch). The controller will hold the drives at their respective ignition positions until the signal to modulate is received from the burner controller. This is performed by closing the circuit on PA10 – PA11 (AUTO input). If the fuel/profile select input is removed (PE9, 10, 11 or 12), the control will immediately move to status 1.
11-14	N/A	NOTE: Statuses 11 to 14 control the light-up sequence of the burner and are provided by the 'burner controller' device used for the application. This control will jump from status 10 to status 15 when the light up is completed. Not Applicable to PPC6000

Stage no.	Stage name	Description
15.	Moving to low fire	Once the AUTO signal is received the fuel and air motors are moved from their ignition positions to their low fire positions. These positions may or may not be the same as the ignition positions. The 'ignition prove' terminal PE7 (low-fire-switch) is switched off. If the fuel/profile select input is removed (PE9, 10, 11 or 12), the control will immediately move to status 1. Once the drives have reached their low fire positions, and an AUTO signal is received (PA10 - PA11), the control will advance to status 16. IF Option 23.0 (warming limit) is enabled, drive will remain at low fire until time set by Option 23.0 expires. NOTE: terminal PE7 will always be OFF during status 15.
16.	Modulation (EK33)	Once the fuel and air motors reach their low fire positions, they are modulated according to the demand placed on the burner. In this burner status the 'high-fire-switch' and 'low-fire-switch' outputs (PE8 & PE7) will come on at high and low fire respectively and may be used for indication purposes. If a 'HIGH' input is provided by closing PA9 – PA11, this control will move to status 17. If the 'AUTO' input is removed by opening PA10 – PA11, this control will modulate down to low fire then jump to status 15. Note: Terminal PE7 will not be energized in this case. This may be used to provide a low fire post purge, or a low fire hold function. If the fuel/profile select input is removed (PE9, 10, 11 or 12), the control will immediately move to status 1. If this control wishes to turn the burner off (there is no call for heat, for example), it will open the 'controlled shutdown' relay, removing power from PE3. It will remain in modulation status however, until one of the conditions above is met.
17.	Move to post-purge	The selected motor(s) are moved to their purge positions all others are moved to their closed positions. When the drives have all stopped, the control will move to status 18.

Stage no.	Stage name	Description
18.	Post-purge	The controller confirms all required drives are at their purge positions, and gives a 'purge proved' signal by providing (line voltage) to terminal PE8 (high-fire-switch). The control will remain in this status until the burner controller signals the end of pre-purge by breaking PA9 – PA11 ('HIGH' input), or the fuel/profile select input is removed (PE9, 10, 11 or 12). In either case, the control will immediately move to status 1.

4.3 Non-volatile lockout

Non Volatile lockouts cannot be cleared without operator intervention and are remembered in the event of power being removed from the control.

A non-volatile lockout will occur under the following conditions:

- In any stage the interface signals are incorrect.
- In stages 5, 7 and 8, stages 10-16 (inclusive) and stage 18 if a motor is not in the correct position
- In any stage, if an internal or external fault not previously mentioned occurs which may affect the safe operation of the burner (see section 0)

4.4 Modulation

During stage 16 (modulation), the control will position the fuel and air motors within the programmed profile appropriate to the requirement for heat. The control has 2 modes of operation using the standard Fireye PID modulation function, Remote and Local. The mode of operation is set via the keypad by pushing the "Burner ON/OFF" key and selecting the mode. Using the programmable block function (option) within the control it is possible for the modulation control to be generated with alternate options, these are not covered in the standard manual as they may be generated by the user. For an overview of Function Block Programming, see Section 10.

4.4.1 Normal/Remote mode.

In Remote mode, the modulation rate is determined by the internal PID control settings, Manual modulation from the keypad, or by one of the following remote influences:

- Auxiliary modulation input,
- Serial communications.

4.4.2 Local mode. (LOC 1, LOC 2)

In Local mode, the modulation rate is determined by either the internal PID settings or Manual modulation via the UP/DOWN keys. External modulation inputs and setpoint selection inputs are ignored.

When "Local1" is displayed the burner is running using the Setpoint 1 PID settings.

When "Local2" is displayed the burner is running using the Setpoint 2 PID settings.

5 Commissioning the control

WARNING

- **This manual may cover more than one model from the PPC6000 series. Check for additional information at the end of this chapter.**
- **While the control is operating in commissioning mode certain safety checks cannot be performed by the control and therefore the safety of the system operation is the sole responsibility of the commissioning engineer.**
- **Do not allow fuel to accumulate in the combustion chamber. If fuel is allowed to enter the chamber for longer than a few seconds without igniting, an explosive mixture could result.**
- **If a flame failure occurs at any point the control will not attempt a re-start until the fault is cleared, unless the option to allow recycling is enabled. Before moving to the ignition position to attempt a re-start the system will perform any selected pre-purge.**
- **Where operating times are adjustable ensure that those selected are acceptable for the appliance being controlled.**
- **Ensure that a purge position is entered for each drive as required, failure to enter a purge position will mean all drives remain at their 'closed' positions.**
- **Once all safety times have been selected it is the responsibility of the commissioning engineer to verify that the times entered are correct for the appliance being controlled.**
- **After entering and/or adjusting any profile points for any profile it is the responsibility of the commissioning engineer to verify that the resulting fuel air ratio is acceptable for the appliance being controlled.**

5.1 General

If any settings in the control are to be changed, it is necessary to enter a commission mode. Three passcodes are available for this purpose.

- *Supplier passcode* - allows entry to all commissioning modes. LV3
- *Adjust Ratio mode*. LV2
- *Site passcode* - allows adjustment of some option parameters. LV1